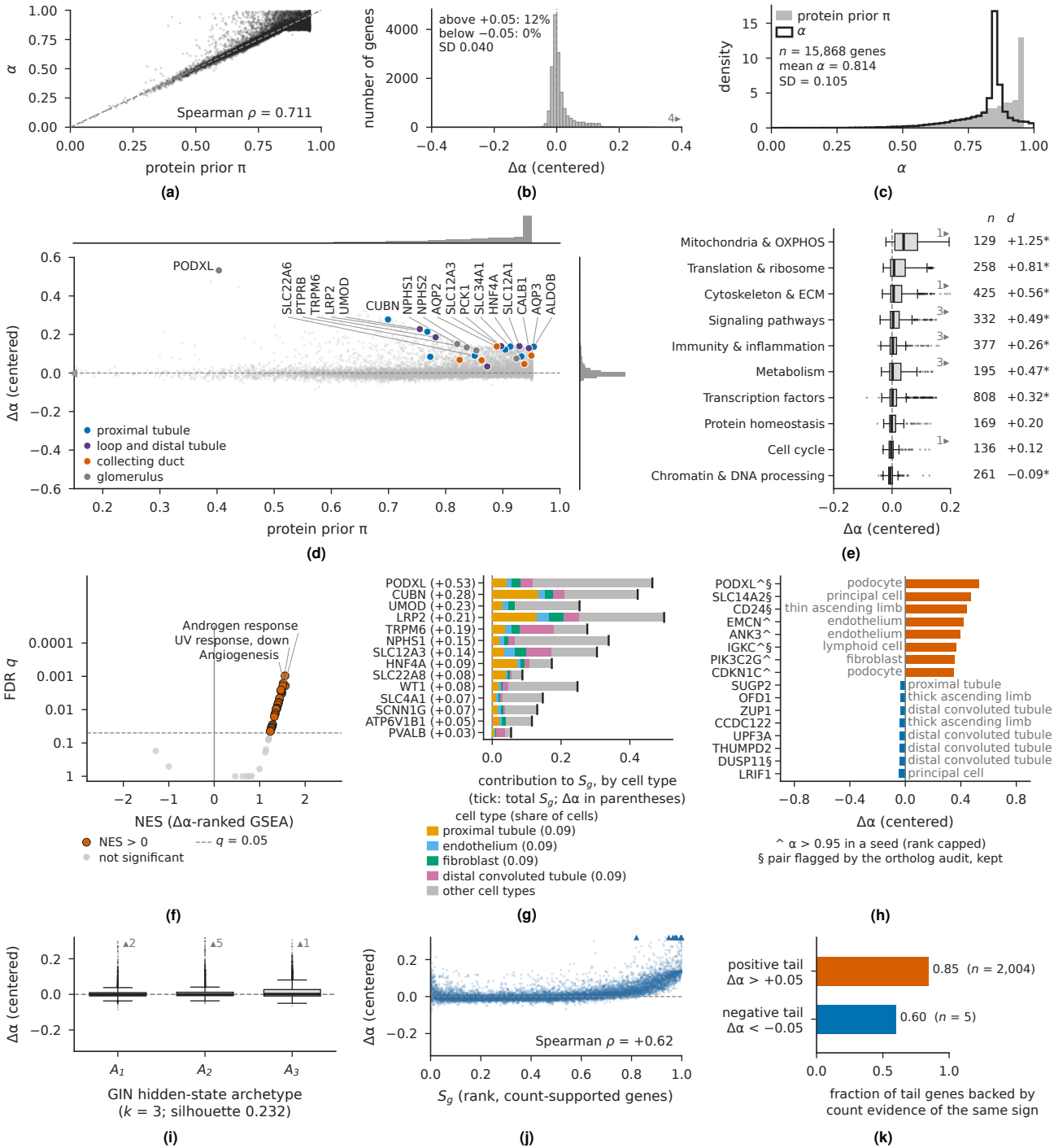

Spearman ρ = 0.711
protein prior π
α
(a)
above +0.05: 12%
below −0.05: 0%
SD 0.040
number of genes
Δα (centered)
(b)
protein prior π
α
n = 15,868 genes
mean α = 0.814
SD = 0.105
density
(c)
PODXL
SLC22A6
PTPRB
TRPM6
LRP2
UMOD
CUBN
NPHS1
NPHS2
AQP2
SLC12A3
PCK1
SLC34A1
HNF4A
SLC12A1
CALB1
AQP3
ALDOB
proximal tubule
loop and distal tubule
collecting duct
glomerulus
Δα (centered)
protein prior π
(d)
n
d
Mitochondria & OXPHOS 129 +1.25*
Translation & ribosome 258 +0.81*
Cytoskeleton & ECM 425 +0.56*
Signaling pathways 332 +0.49*
Immunity & inflammation 377 +0.26*
Metabolism 195 +0.47*
Transcription factors 808 +0.32*
Protein homeostasis 169 +0.20
Cell cycle 136 +0.12
Chromatin & DNA processing 261 −0.09*
Δα (centered)
(e)
Androgen response
UV response, down
Angiogenesis
FDR q
NES (Δα-ranked GSEA)
NES > 0
not significant
q = 0.05
(f)
PODXL (+0.53)
CUBN (+0.28)
UMOD (+0.23)
LRP2 (+0.21)
TRPM6 (+0.19)
NPHS1 (+0.15)
SLC12A3 (+0.14)
HNF4A (+0.09)
SLC22A8 (+0.08)
WT1 (+0.08)
SLC4A1 (+0.07)
SCNN1G (+0.07)
ATP6V1B1 (+0.05)
PVALB (+0.03)
contribution to S_g, by cell type
(tick: total S_g; Δα in parentheses)
cell type (share of cells)
proximal tubule (0.09)
endothelium (0.09)
fibroblast (0.09)
distal convoluted tubule (0.09)
other cell types
(g)
PODXL^§ podocyte
SLC14A2§ principal cell
CD24§ thin ascending limb
EMCN^ endothelium
ANK3^ endothelium
IGKC^§ lymphoid cell
PIK3C2G^ fibroblast
CDKN1C^ podocyte
SUGP2 proximal tubule
OFD1 thick ascending limb
ZUP1 distal convoluted tubule
CCDC122 thick ascending limb
UPF3A distal convoluted tubule
THUMPD2 distal convoluted tubule
DUSP11§ distal convoluted tubule
LRIF1 principal cell
Δα (centered)
^ α > 0.95 in a seed (rank capped)
§ pair flagged by the ortholog audit, kept
(h)
Δα (centered)
GIN hidden-state archetype
(k = 3; silhouette 0.232)
(i)
Δα (centered)
Spearman ρ = +0.62
S_g (rank, count-supported genes)
(j)
positive tail
Δα > +0.05
0.85 (n = 2,004)
negative tail
Δα < −0.05
0.60 (n = 5)
fraction of tail genes backed by
count evidence of the same sign
(k)